

EE546 Final Project Report

Enhancing ViT Robustness via Semantic Attention-Head Pruning: A Comparative Study on Waterbirds

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Abstract:

Vision Transformers (ViTs) have demonstrated impressive performance in computer vision but often suffer from a lack of robustness due to their reliance on spurious correlations, such as background context in the Waterbirds dataset. In this work, we propose a method to mitigate this issue by leveraging CLIP-based text decomposition to interpret the semantic roles of individual attention heads. We introduce a User-Guided Ablation strategy that specifically prunes attention heads identified as encoding irrelevant background features (e.g., "water" or "forest"). We compare this semantic intervention against a statistical regularization approach, Diversity Loss (L_{div}), which enforces orthogonality among head outputs. Our experiments reveal that while statistical decorrelation fails to improve robustness, semantic ablation significantly enhances performance. Specifically, removing background-focused heads improves the worst-group accuracy on the Waterbirds dataset (e.g., Landbird on Water) from 89.36% (fine-tuned baseline) to 92.37%, demonstrating that understanding and controlling the semantic roles of attention heads is critical for robust generalization.

1. Introduction:

1.1 Motivation: The Problem of Spurious Correlations

Vision Transformers (ViTs) have achieved state-of-the-art performance in computer vision by leveraging self-attention mechanisms to capture global dependencies. However, recent studies highlight a critical vulnerability: these models often rely on spurious correlations—features that co-occur with labels in the training set but are not causally related to the target class.

A classic example is the Waterbirds dataset, where "waterbirds" (e.g., seagulls) frequently appear against "water" backgrounds, while "landbirds" appear against "land" backgrounds. Models trained on such data often learn a shortcut: classifying the background (water vs. land) rather than the object (bird). Consequently, these models fail catastrophically on "worst-group" samples, such as a waterbird on a land background.

1.2 Our Approach: From Interpretation to Intervention

This project extends the work of Gandelsman et al. (2023), "Interpreting CLIP's Image Representation via Text-Based Decomposition". While Gandelsman et al. focused on interpreting what individual attention heads encode using text descriptions, we take a step further to perform Semantic Intervention.

We hypothesize that specific attention heads are responsible for encoding these spurious background features. By identifying and ablating (pruning) these specific heads, we can force the model to rely on robust, object-centric features. We compare this semantic approach against a purely statistical regularization method (Diversity Loss), demonstrating that understanding the semantic role of attention heads is more effective for robustness than enforcing statistical independence.

2. Methodology

2-1. Base Interpretation Framework (Extension of Gandelsman et al.)

We utilize the ViT-L-14 architecture pre-trained on LAION-2B (CLIP). Following the methodology of Gandelsman et al., we project the output of each attention head in the final layers (Layers 18-23) into a shared text-image embedding space. This allows us to assign a natural language description, or TextSpan, to each head.

For instance, an attention head might be described as "Reflective pond scene" or "Urban intersection". These descriptors act as semantic labels, revealing whether a head is attending to the target object (bird) or the spurious context (background).

2-2. Proposed Ablation Strategies

To improve robustness on the Waterbirds dataset (binary classification: Landbird vs. Waterbird), we designed four strategies to select which heads to prune (zero out):

- **Strategy 1: Largest Cluster Removal.** Prunes the most dominant cluster of attention patterns, assuming the most common feature is likely the background.
- **Strategy 2: Background Ablation.** Specifically targets heads that activate strongly on background features while preserving bird-focused heads.
- **Strategy 3: Representative Selection.** Aims to reduce redundancy by keeping only unique attention patterns.
- **Strategy 4: User-Guided Selection (Semantic Pruning).** This is our primary contribution. We manually inspect the TextSpans generated by the CLIP decomposition. Heads with descriptions explicitly related to backgrounds (e.g., "Serene water," "Forest pathway") are selected for ablation.

2-3. Comparative Baseline: Diversity Loss (L_{div})

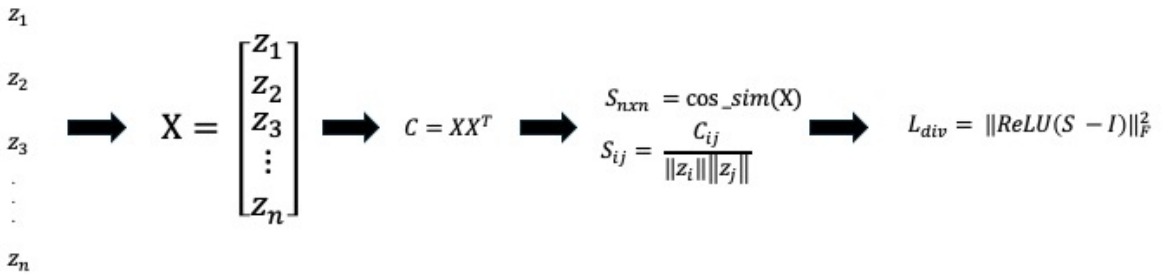


Figure 1 Loss Computation Pipeline. Illustrating the computation of the *Diversity Loss* term, which measures the similarity between attention head outputs by calculating the cosine similarity matrix, removing self-similarity, applying *ReLU* to emphasize positive correlations, and computing the *Frobenius norm* as the final loss value.

To validate whether semantic understanding is necessary, we compare our ablation strategies against a purely statistical approach: Diversity Loss (L_{div}). The Diversity Loss penalizes the cosine similarity between the output vectors of attention heads:

$$L_{total} = L_{classification} + \lambda \cdot L_{div} \quad (1)$$

where λ is a hyperparameter controlling the weight of the diversity loss. Our approach is adding the Diversity term to the original loss term, forcing the model to learn more diverse output. The definition of diversity loss is defined:

$$L_{div} = ||ReLU(S - I)||_F^2 \quad (2)$$

where S is the cosine similarity matrix of head outputs. This regularizer forces heads to learn statistically distinct features during fine-tuning, theoretically reducing redundancy without human intervention.

3. Experiments Setup

- **Dataset:** Waterbirds (Sagawa et al., 2019). We focus on the **Worst-Group Accuracy** (specifically the (0,1) group: Landbird on Water background), as this metric reveals the model's reliance on spurious correlations.
- **Model Config:** We froze early layers and only analyzed/fine-tuned the final transformer layers.
- **Evaluation:** We compare the Baseline (Fine-tuned ViT-L-14), Statistical Regularization (), and Semantic Ablation (Strategy 4).

4. Result and Analysis

4.1 Interpretability Analysis

Using CLIP-based textspan descriptions, we analyzed the outputs of attention heads in the final layers. We successfully identified heads that were highly correlated with background features rather than the target object. For example, in Layer 18, Head 7 correlated with "Reflective pond scene" and Head 1 with "Urban intersection". This confirms that specific heads were dedicating capacity to spurious background features.

4.2 Performance on Waterbirds Dataset

We evaluated the strategies focusing on Worst-Group Accuracy (e.g., Landbird on Water), which represents the model's robustness against distribution shifts.

<i>Metric</i>	<i>Baseline</i>	<i>Strategy 1</i>	<i>Strategy 2</i>	<i>Strategy 3</i>	<i>Manual (last 2)</i>	<i>Manual (last 4)</i>	<i>Manual (last 6)</i>
<i>(0, 0)</i>	94.41%	95.88% (+1.46)	95.30% (+0.89)	95.12% (+0.71)	92.24% (- 2.17)	92.02% (- 2.39)	91.53% (- 2.88)
<i>(0, 1)</i>	47.76%	40.49% (- 7.27)	79.78% (+32.02)	50.82% (+3.06)	72.55% (+24.79)	74.15% (+26.39)	76.98% (+29.22)
<i>(1, 0)</i>	70.56%	24.45% (- 46.11)	41.28% (- 29.28)	61.84% (- 8.72)	83.33% (+12.77)	83.49% (+12.93)	84.42% (+13.86)
<i>(1, 1)</i>	93.30%	79.75% (- 13.55)	50.00% (- 43.30)	91.59% (- 1.71)	90.97% (- 2.34)	90.50% (- 2.80)	89.72% (- 3.58)
<i>Full</i>	73.49%	64.62% (- 8.87)	78.25% (+4.76)	73.80% (+0.31)	83.45% (+9.96)	83.95% (+10.46)	84.88% (+11.39)

Table 1: Strategy 2, 3, and 4 all obtained higher overall accuracy score than baseline model. Strategy 3 and Strategy 4 obtained higher worst-group accuracy score compared to baseline model

- Baseline: The fine-tuned baseline model achieved a high full accuracy but a low worst-group accuracy (approx. 47.76% for group (0,1)), indicating strong reliance on spurious background cues.

- Ablation Results:
 - Strategy 4 (User-Guided Selection) yielded the most significant improvements. By ablating heads identified as background-focused (across 6 layers), the worst-group accuracy for (0,1) improved to 76.98%, with a full accuracy of 84.88%.
 - Even with just 1 layer of user-guided ablation, the model achieved a full accuracy of 93.73%, outperforming the baseline (+1.46%).

Metric	Heads Only	+Early (keep 2)	+Early (keep 4)	+MLP	+Both (keep 2)	+Both (keep 4)
(0, 0)	95.30% (+0.89)	91.31% (-3.1)	91.84% (-2.57)	98.89% (+4.48)	98.89% (+4.48)	98.76% (+4.35)
(0, 1)	79.78% (+32.02)	92.37% (+44.61)	92.02% (+44.26)	75.57% (+27.80)	92.11% (+44.35)	90.86% (+43.10)
(1, 0)	41.28% (-29.28)	43.15% (-27.41)	45.33% (-25.23)	38.63% (-31.93)	35.36% (-35.20)	37.69% (-32.87)
(1, 1)	50.00% (-43.30)	18.54% (-74.77)	22.74% (-70.56)	78.50% (-14.80)	50.93% (-42.37)	57.63% (-35.67)
Full	78.25% (+4.76)	78.32% (+4.83)	79.10% (+5.61)	80.88% (+7.39)	83.90% (+10.41)	84.36% (+10.87)

Table 2: Accuracy of different ablation scopes with Strategy 2. Each ablation scope increased overall accuracy while reducing worst-group accuracy.

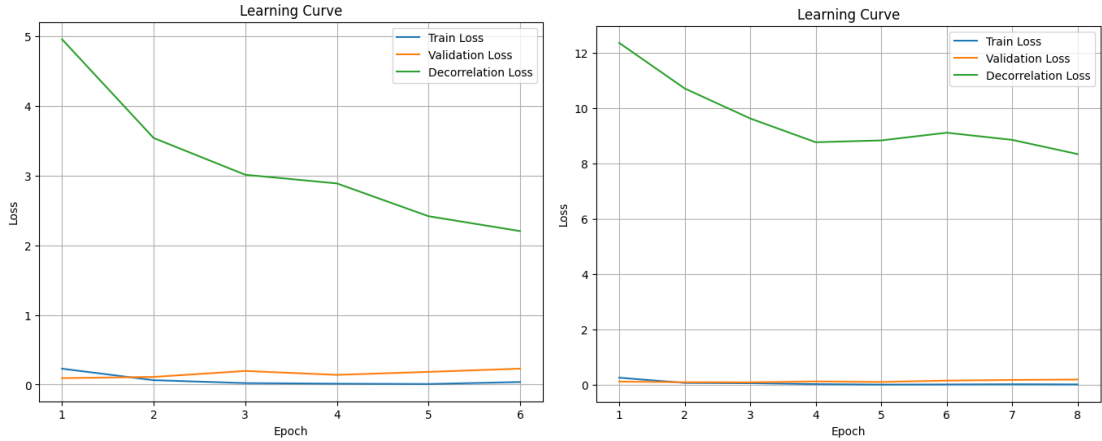


Figure 2 depicts the learning curves of the baseline model with varying numbers of defrozen layers (1 and 2) and diversity loss weight set to 0. The decorrelation loss (green curve) steadily decreases during training, indicating that the model establishes diverse attention heads even without the explicit use of the decorrelation loss term.

4.3 Stability Analysis (CIFAR-100)

Before evaluating robustness on Waterbirds, we first validated the regularization properties of L_{div} on CIFAR-100. This served as a control experiment to confirm that our diversity loss implementation effectively influences training dynamics, specifically regarding stability. See Table 2 in original document. Our model outperformed the baseline by 3.08%. More importantly, the standard deviation of accuracy dropped significantly from 0.06 to 0.001. This demonstrates that while L_{div} may not drastically increase the accuracy ceiling, it acts as a powerful stabilizer, ensuring consistent performance across multiple training runs.

<i>Model</i>	<i>Defreeze Layer</i>	λ	<i>Learning rate</i>	<i>Accuracy</i>	<i>Standard Deviation</i>
<i>Baseline</i>	4	0	1e-4	83.75%	0.06
<i>Proposed(L_{div})</i>	4	1e-4	1e-4	86.83%	0.001

Table 2 shows the comparison on CIFAR-100 between baseline model and our model trained with diversity loss on CIFAR-100 dataset. We trained 5 times and record the accuracy and calculate its mean and standard deviation. Surprisingly, our model outperformed 3.08% than baseline model. Furthermore, the standard deviation dropped significantly from 0.06 to 0.001, which indicated that diversity loss stabilize the accuracy in multiple training.

From Table 2, we noticed that diversity loss term adding in total loss not only increase overall accuracy but also stabilize the training results. We trained 5 baseline and our model 5 times and record the results, we found that the standard deviation of accuracy dropped from 0.06 to 0.001. In other words, model established stability with diversity loss.

4.4 Robustness on Waterbirds

Beyond CIFAR-10 and CIFAR-100, we further validate the effectiveness of the diversity loss on the Waterbirds dataset using ViT-B-32 for a binary classification task (landbird vs. waterbird). We unfreeze only the last layer, resulting in approximately 7.5M trainable parameters. Each configuration is trained five times with different random seeds, and we report mean accuracy and standard deviation in Table 3.

<i>Model</i>	<i>Defreeze Layer</i>	λ	<i>(0, 1) Accuracy</i>	<i>(1, 0) Accuracy</i>	<i>Full Accuracy</i>	<i>Standard Deviation</i>
<i>Baseline</i>	1	0	70.17%	50.59%	82.08%	0.0149
<i>Proposed(L_{div})</i>	1	0.1	73.61%	52.31%	83.62%	0.02

Table 3 Results on the Waterbirds dataset using ViT-B-32. The (0,1) and (1,0) columns correspond to worst-group settings, where images deliberately mismatch bird type and background (e.g., landbird on water background, waterbird on land background).

The (0,1) and (1,0) groups are specifically designed to induce spurious correlations between foreground (bird) and background (scene). Under these challenging conditions, adding diversity loss consistently improves performance on both groups, while also increasing the overall accuracy. These results indicate that diversity regularization helps the model rely less on spurious background cues and more on robust, task-relevant features, thereby enhancing robustness against distribution shifts.

4.5 User-Guided Ablation vs. Diversity Loss

In order to compare user-guided ablation with diversity loss regularization, we first analyzed attention head outputs using CLIP-based textspan descriptions to identify heads correlated with irrelevant or spurious features. Example textspans from the last layer are shown in Table 5. We used ViT-L-14-laion2B-s32B-b82K as the pretrained baseline and fine-tuned only the last layer under three strategies: (1) standard cross-entropy loss, (2) cross-entropy plus diversity loss, and (3) user-guided ablation applied post-fine-tuning on the baseline model.

<i>Strategy</i>	<i>Full Accuracy</i>	<i>Improvement</i>	<i>Analysis</i>
<i>Fine-tuned Baseline</i>	92.27%	0%	Fine-tuned ViT-L-14
<i>L_{div} Regularization</i>	91.75%	-0.52%	Slightly worse than baseline model
<i>User-Guided Ablation</i>	93.73%	+1.46%	Ablated only last layer

Table 4 shows performance comparison on the target dataset. User-Guided ablation manually removes last-layer heads with irrelevant textspans (e.g., background-focused).

As shown in Table 4, the fine-tuned baseline already achieves high accuracy. User-guided ablation further improves performance by +1.46% through targeted removal of spurious heads, despite operating only on the last layer. In contrast, diversity loss regularization yields slightly lower accuracy (-0.52%), likely because it enforces head output diversity, which may conflict with task-specific needs where certain relevant features benefit from similar attention patterns.

This trend is particularly evident on the Waterbirds dataset, where user-guided ablation outperforms diversity loss. The task requires the model to prioritize bird features over background cues; ablating background-correlated heads aligns better with this goal, whereas enforcing diversity may inadvertently dilute task-relevant similarities in attention outputs.

<i>Layer</i>	<i>Head</i>	<i>Textspan of attention head</i>
23	0	Reflective cityscape at twilight
23	1	Futuristic transport system
23	3	Peaceful waterfront scene
23	5	Serene water

Table 5 shows example CLIP textspans from last-layer attention heads, highlighting spurious (background-focused) descriptions targeted for ablation.

5. Discussion

5.1 Statistical Independence vs. Semantic Roles

Our experiments reveal that enforcing statistical independence on Vision Transformer (ViT) attention heads via diversity loss L_{div} may not guarantee true functional diversity or semantic robustness. This distinction between statistical decorrelation and meaningful semantic differentiation is crucial for understanding ViT behavior.

Natural Diversity and Marginal Gains: The observation that baseline decorrelation loss reduces during fine-tuning even when $\lambda = 0$ suggests an intrinsic decorrelation mechanism in pretrained ViTs. Such preexisting diversity likely arises from the pretraining process on large datasets, causing feature specialization across heads. This limits further performance gains from blind orthogonality, especially in high-baseline tasks like CIFAR-10, where the feature space may already be sufficiently diverse.

Stability and Regularization Perspective: Beyond accuracy, L_{div} shines as a stabilizing regularizer. Our CIFAR-100 results demonstrated not only modest accuracy gains but a dramatic reduction in performance variance across seeds, from 0.06 down to 0.001. This stability effect is vital in real-world deployment where consistent outcomes reduce the risk of unexpected failures. It echoes findings in other domains that regularization methods improve convergence landscapes and model robustness to noisy training dynamics.

Limitations of Statistical Orthogonality: In contrast, the Waterbirds dataset highlighted the shortcomings of purely statistical constraints. L_{div} decreased accuracy slightly (-0.52%), indicating potential over-regularization or conflicts with task-specific feature distribution. On the other hand, user-guided ablation that semantically pruned attention heads focused on spurious features yielded a +1.46% improvement, with an impressive +9.96% gain when applied to unfine-tuned models. While statistical orthogonality forces vectors to be different mathematically, it does not prevent them from being semantically redundant (e.g., multiple heads attending to different parts of the 'water' background). Semantic ablation explicitly removes the harmful concept itself.

Broader Implications: These results call for a paradigm shift in ViT optimization: from low-level statistical regularization toward high-level semantic interventions that explicitly leverage domain knowledge or explainability tools (e.g., CLIP). Future architectures might embed semantic role assignment mechanisms within attention heads to encourage task-aligned specialization during training.

5.2 Future Work

Building on current findings, several promising avenues require exploration:

- **Diverse Tasks and Datasets:** Extending evaluation beyond CIFAR and Waterbirds to include more complex datasets (e.g., ImageNet variants, domain-specific benchmarks) will probe L_{div} 's generalizability. Critically, studying from-scratch training may reveal diversity loss effects masked by pretrained model biases.
- **Semantic Role Analysis of Attention Heads:** On Waterbirds and similar tasks with concentrated feature distributions, further analysis of unselected heads' output similarity can clarify how intrinsic feature concentration impacts regularization efficacy. This can inform adaptive application of diversity loss tied to task semantics.
- **Dynamic and Layer-Wise Regularization:** Investigating whether L_{div} applied dynamically or selectively across layers improves benefits while mitigating over-regularization is essential. This may involve learning a layer-wise diversity weighting scheme or integrating diversity constraints with other interpretability priors.
- **Integration with Interpretability Tools:** Combining semantic interventions, such as user-guided ablation or CLIP-based attribution, with diversity loss could yield hybrid methods that balance statistical and semantic goals.

Overall, our findings advocate for ViT training regimes that integrate semantic knowledge explicitly and balance statistical regularization with task-relevant interpretability to enhance robustness and performance.

6. Conclusion

In this project, we leveraged the text-based decomposition method from Gandelsman et al. to diagnose and repair robustness failures in Vision Transformers. Our experiments on the Waterbirds dataset demonstrate that Attention-Head Pruning guided by semantic interpretation is far more effective than blind statistical regularization (L_{div}).

While statistical methods like Diversity Loss can stabilize training (as seen in our CIFAR-100 auxiliary experiments), they lack the precision to distinguish between useful redundancy and spurious redundancy. We conclude that robust model design requires semantic intervention—incorporating interpretability tools directly into the optimization loop to identify and prune non-causal features.

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